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To the Editors
Expert Systems with Applications

Dear Editors,

Please consider our manuscript "Reinforcement Learning vs. Rule-Based Policies for Dynamic Credit Threshold Control: A Simulation Study" for publication as a Full Length Article.

Consumer lenders do not set an acceptance threshold once. They revise it as the portfolio composition, the capital usage and the market conditions drift, which makes the accept/reject cut-off a recurring control decision rather than a one-shot classification step. The credit-scoring literature still largely optimises a static cut-off and scores it by AUC or Gini, which leaves the operational question open: is it worth automating weekly threshold control with a learned policy, and under what conditions?

We answer that question with a controlled experiment. Six deep reinforcement learning agents (DQN, Double-DQN, PPO, SAC, A3C, A2C) compete against five economically motivated rule-based policies in a calibrated credit-portfolio simulator, across six market scenarios and four observation sizes (12D, 20D, 30D, 50D), with 1,500 training episodes per seed, eight seeds, and paired statistical tests on run-level data. Only the observation dimensionality changes between conditions; the simulator, the reward, the seeds and the evaluation protocol are fixed and verified programmatically.

The answer is conditional, and part of it is negative. Rule-based weekly search wins the reward ranking in every scenario and every dimension, and it needs no training, so it stays the safe default when training resources are limited. The large RL deficit we measured under a short budget, however, turns out to be an artefact of that budget rather than a ceiling: at 1,500 episodes the profit gap narrows to between 0.46% and 2.52%, and at 30D and 50D the best RL agent holds lower default rates than both profit-seeking baselines in five of six scenarios. A lender who values capital conservation can trade a small profit reduction for that risk-side gain, and the paper states when the trade is available.

What is new here relative to the published literature. The closest precursor, Herasymovych et al. (2019), trained a DQN agent to move a single shared acceptance threshold against one static baseline in one scenario. We split the threshold by borrower segment (new and repeat clients), vary the observation space from 12D to 50D under a protocol that is otherwise fixed and machine-verified, compare six RL algorithms against five rule-based policies rather than one, run six market scenarios, and test each claim on paired run-level data. Kiatsupaibul et al. (2024) survey reinforcement learning for credit scoring and underwriting and note that controlled comparisons against strong rule-based policies are largely missing; this paper supplies one. The work also continues the credit-risk line published in Expert Systems with Applications (Bellotti and Crook, 2009; Brown and Mues, 2012; Xia et al., 2017; Zhang and Yu, 2023; Yang et al., 2024), and differs from it in treating the decision as sequential control under delayed outcomes rather than as a static cut-off, which is the comparison a lender faces in practice.

One word on the environment. Loan-level panels that record the score, the funding

decision, the realised outcome and the calendar timing across several market regimes are proprietary, so a calibrated simulator is the only way to vary one factor at a time under a fixed protocol. Every primary parameter is calibrated to published retail lending statistics, the score-to-default relationship is compared against the Lending Club loan book, and Section 7 sets out the limits of the design. We read the results as evidence about controller behaviour under controlled conditions, not as a forecast for any specific portfolio.

The manuscript is original, has not been published before, and is not under consideration elsewhere. Both authors approved the submitted version. We declare no competing interests. Following the journal's double-anonymized policy, the manuscript file carries no names, affiliations or acknowledgements; these are in the separate title page.

We hope the paper is a fit for Expert Systems with Applications.

Sincerely,
Evgeny Baulin, on behalf of the authors
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